

Hysteresis Analysis

Effect of MD1 Conditioning on SFTPRO & LHC Cycles

MBB max-speed + NMR campaign — 2026-03-06

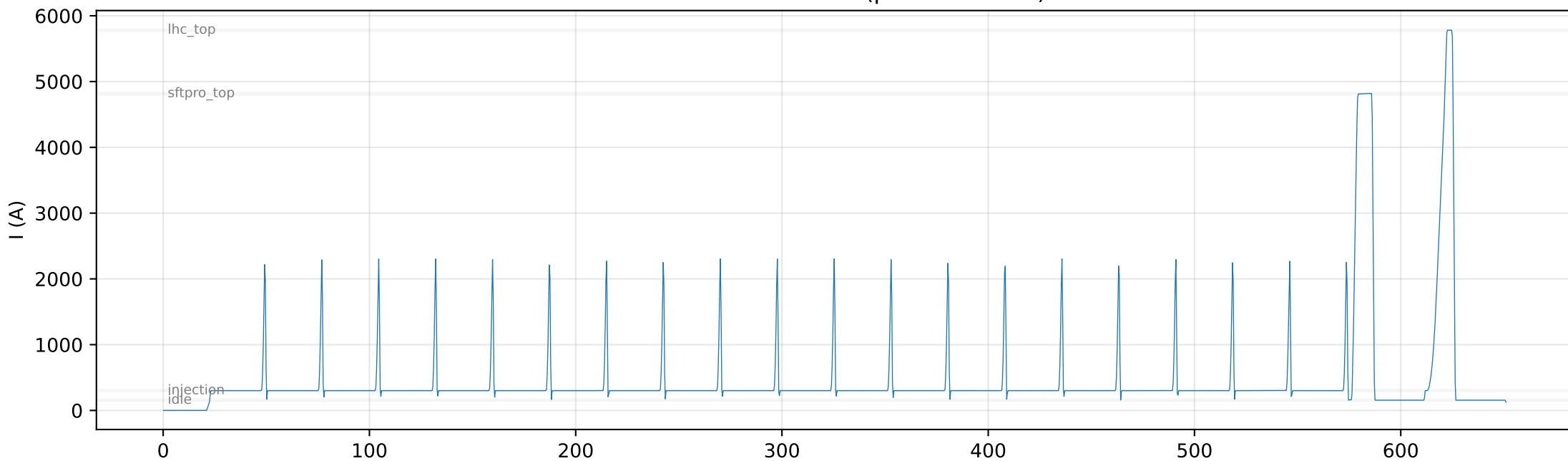
200 GeV session: 20× full MD1 (peak ~2267 A) → SFTPRO → LHC

26 GeV session: 20× flat MD1 (peak ~301 A) → SFTPRO → LHC

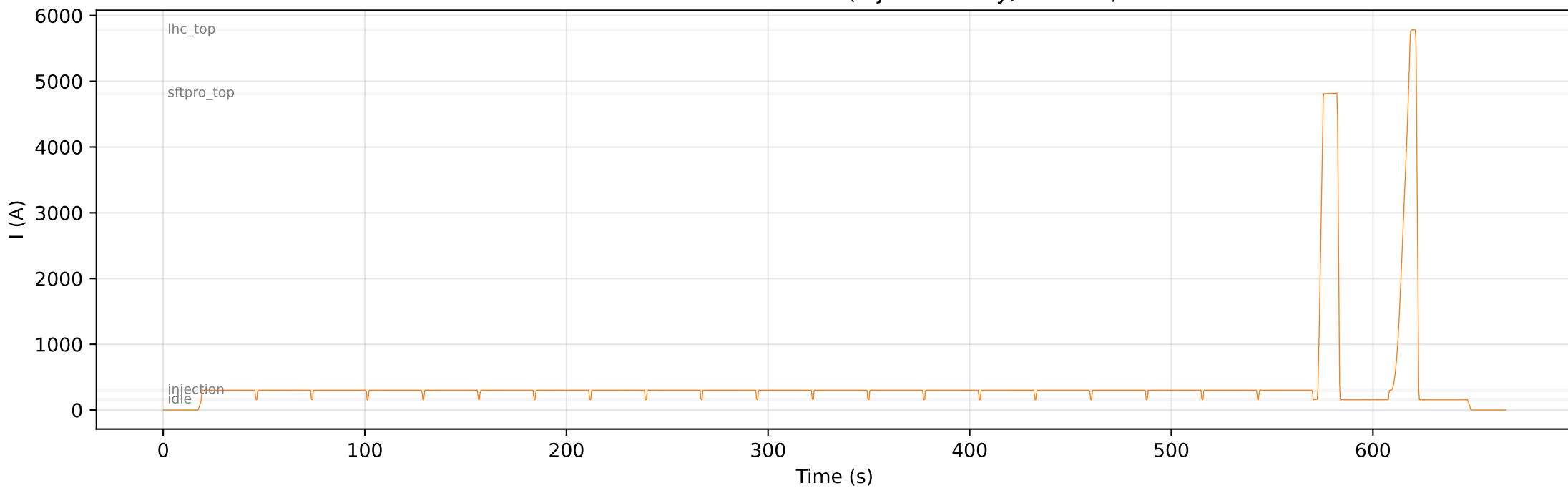
Body segment • Encoder offset corrected • Plateau by $|\text{ramp rate}| < 5 \text{ A/s}$

Current Profile — Both Sessions

200 GeV — Full MD1 (peak ~2267 A)

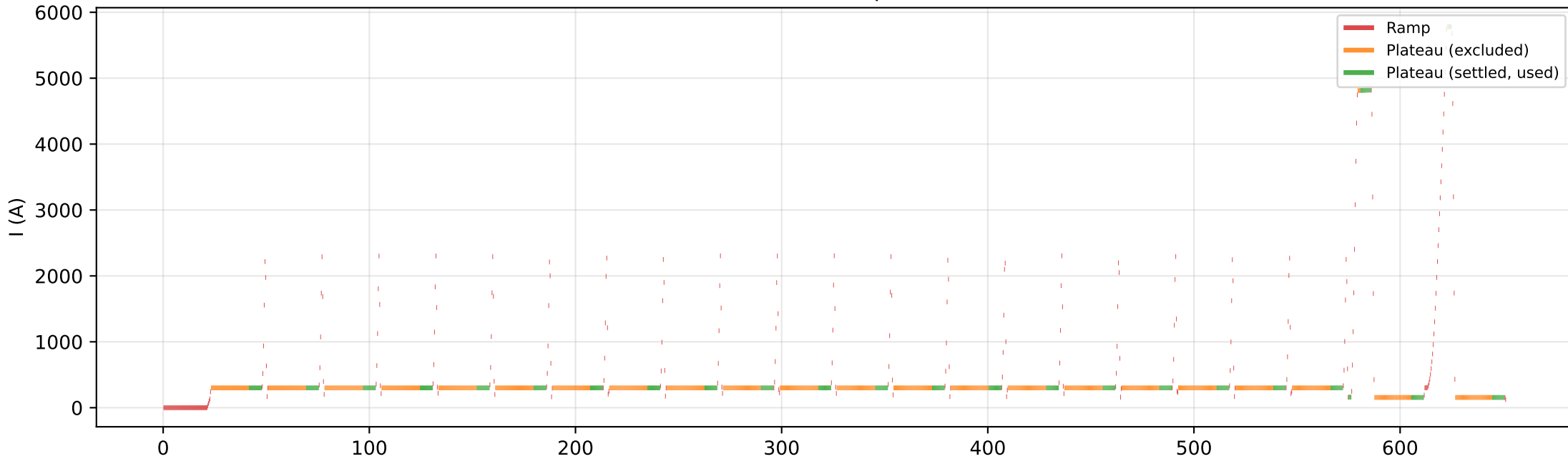


26 GeV — Flattened MD1 (injection only, ~301 A)

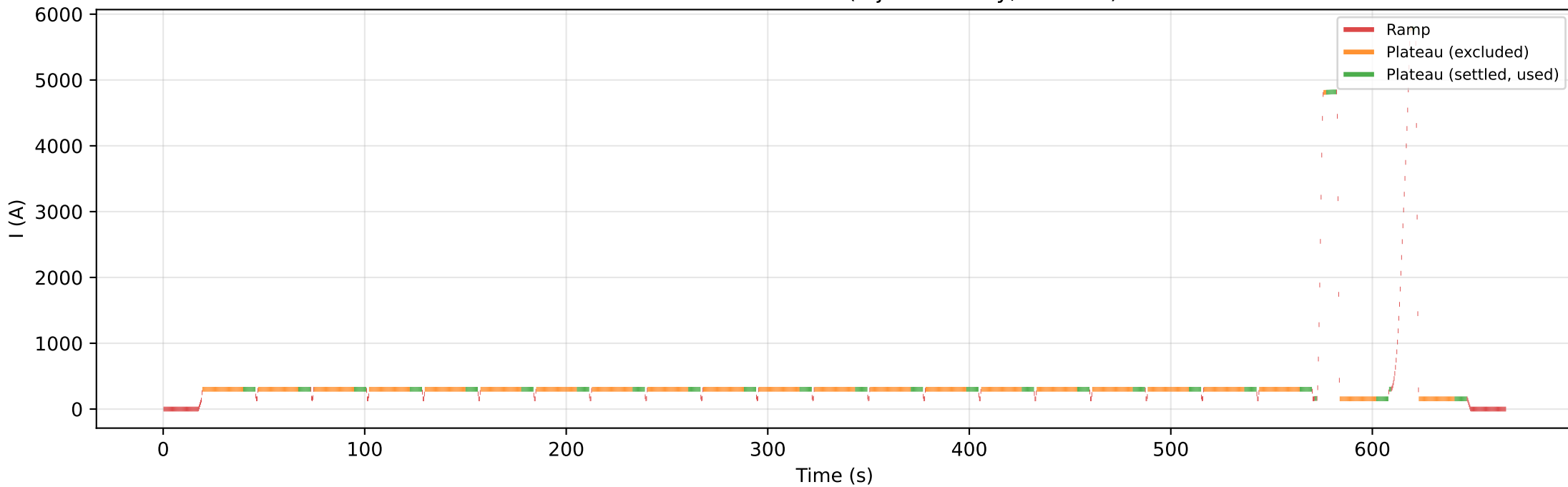


Current Profile with Turn Classification

200 GeV — Full MD1 (peak ~2267 A)

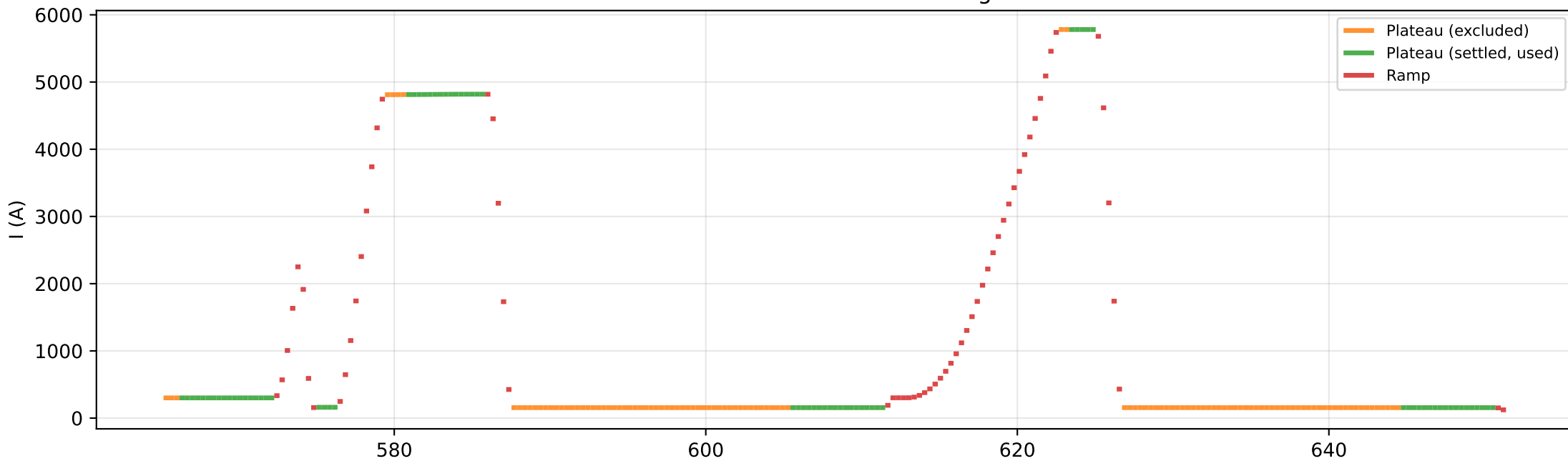


26 GeV — Flattened MD1 (injection only, ~301 A)

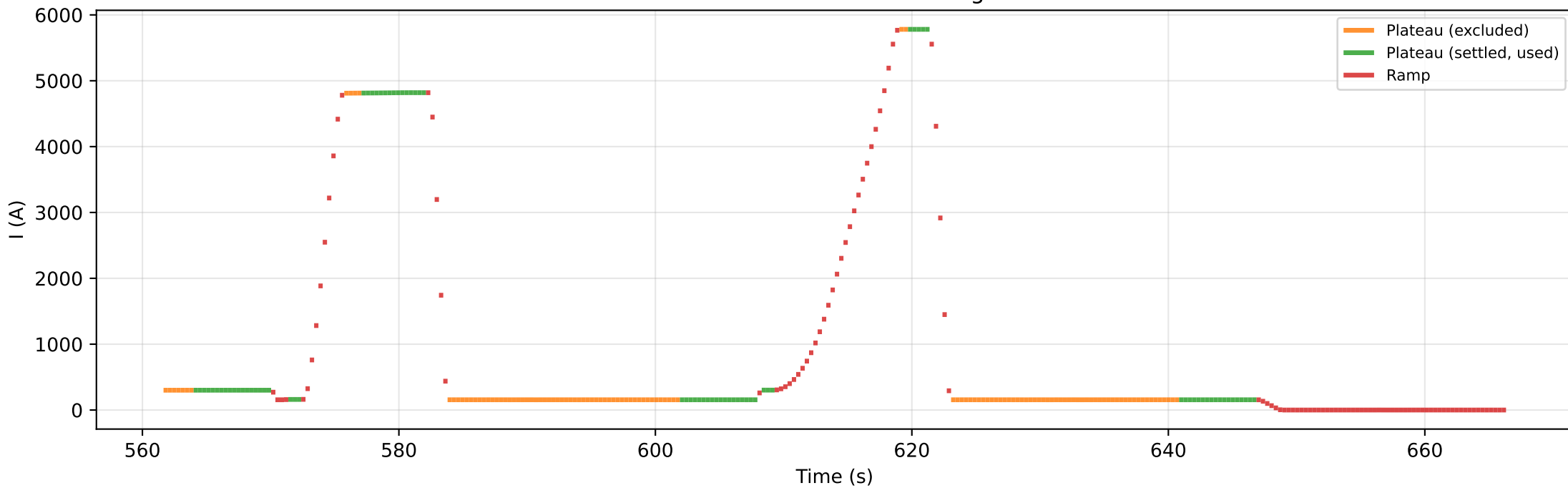


SFTPRO + LHC Region — Turn Classification (zoom)

200 GeV — SFTPRO + LHC region

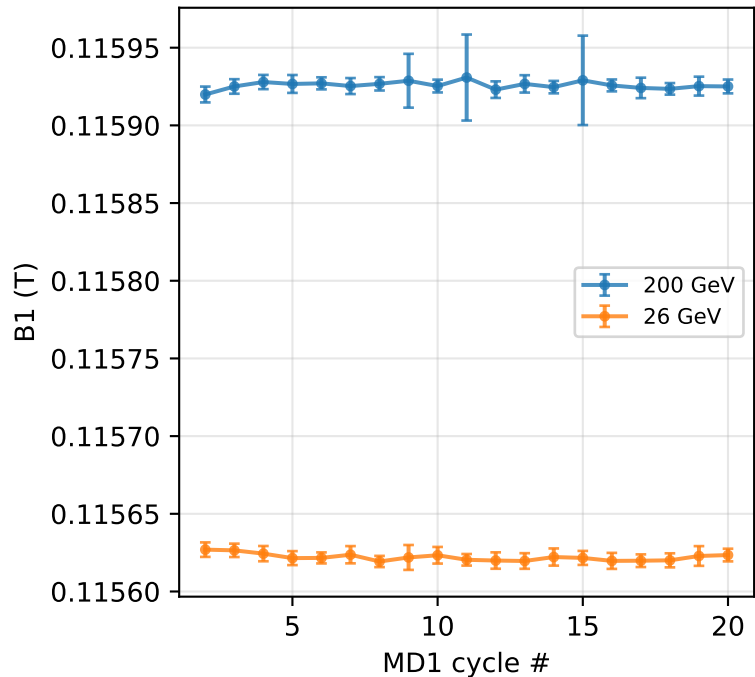


26 GeV — SFTPRO + LHC region

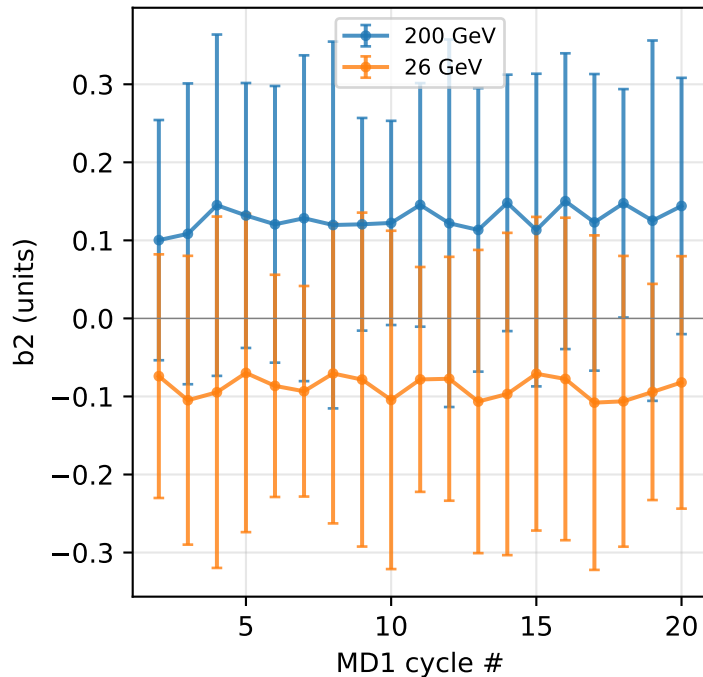


MD1 Minor Loop Convergence — Injection Plateau (body, cycle 1 excluded)

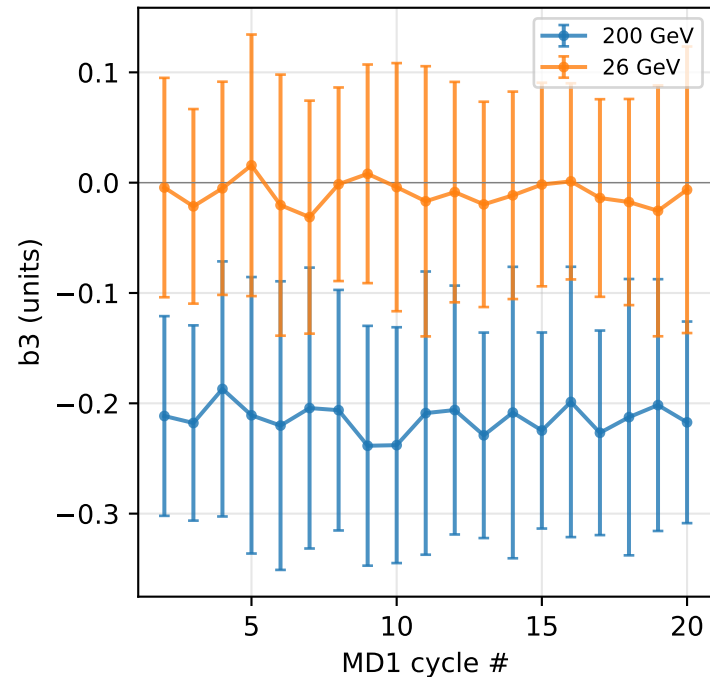
B1 (T) at injection (~301 A)



b2 (units) at injection (~301 A)

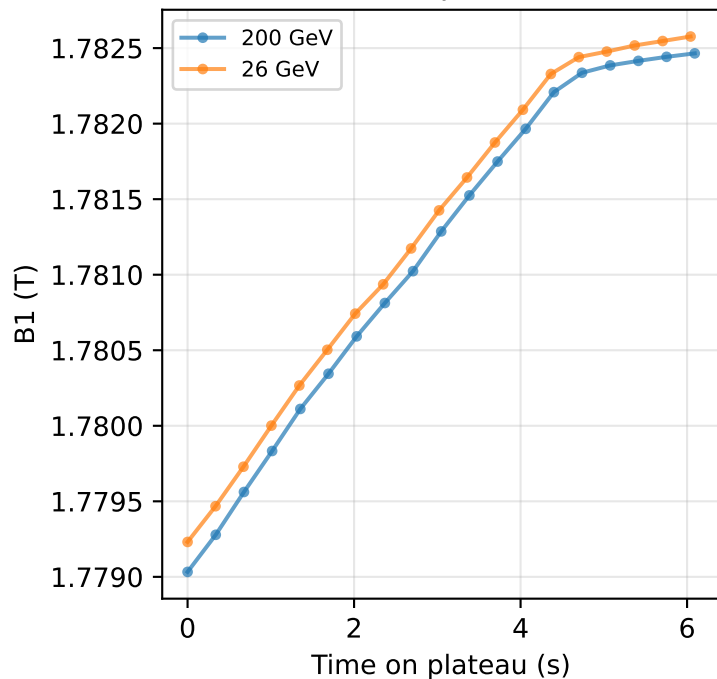


b3 (units) at injection (~301 A)

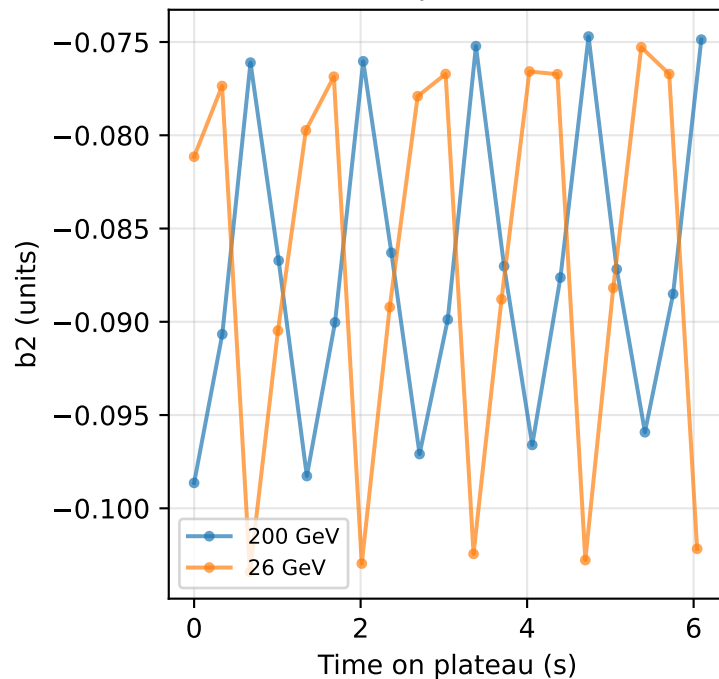


SFTPRO Top Plateau — Turn-by-Turn (body)

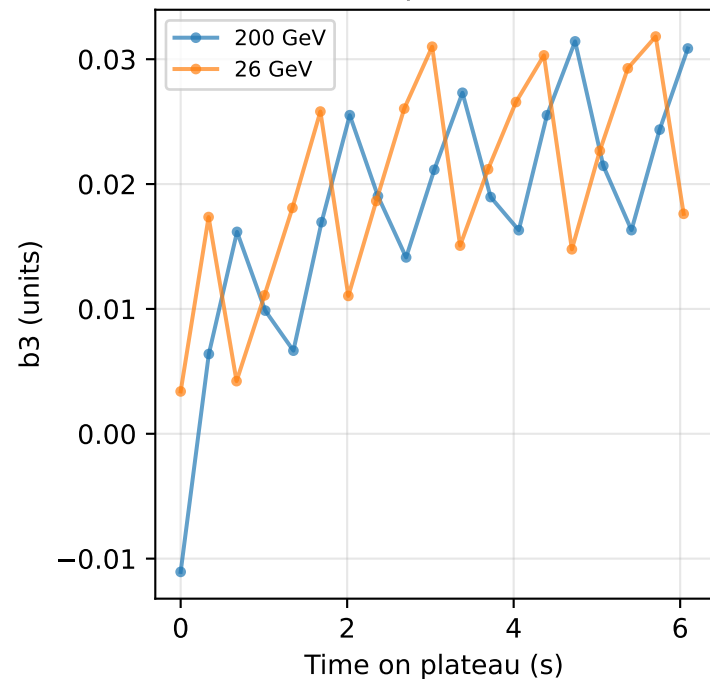
SFTPRO top — B1 (T)



SFTPRO top — b2 (units)

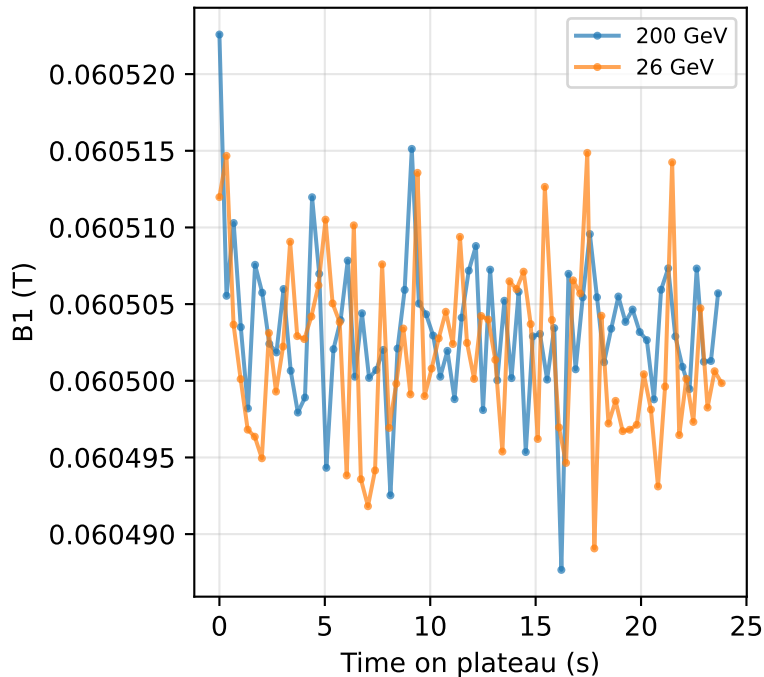


SFTPRO top — b3 (units)

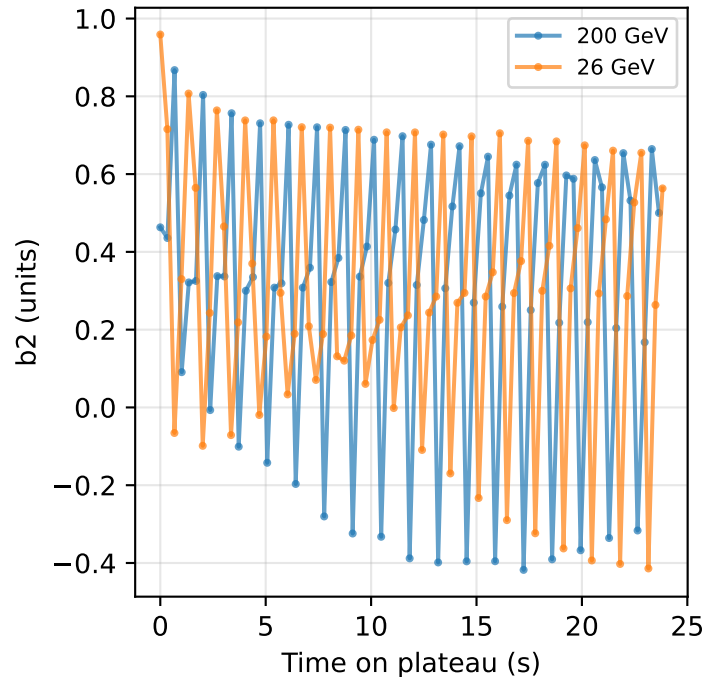


Post-SFTPRO Idle Plateau — Turn-by-Turn (body)

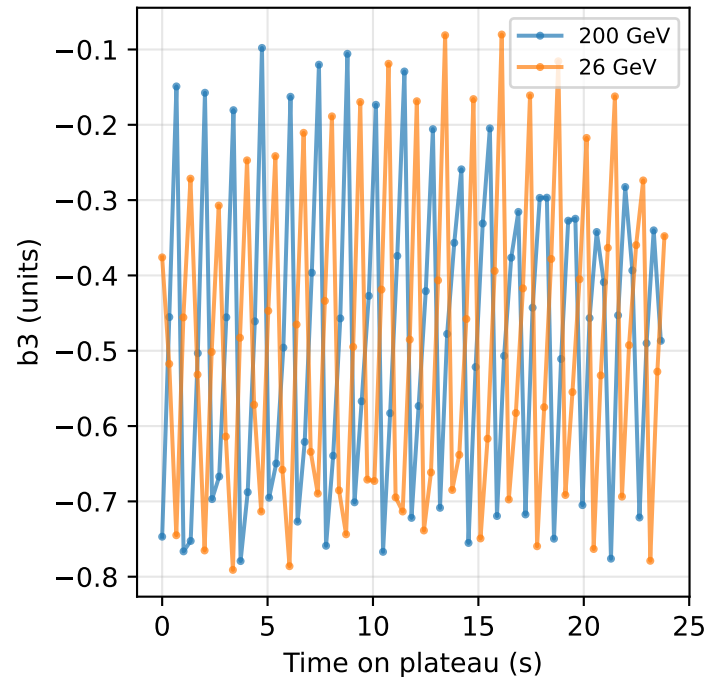
Post-SFTPRO idle — B1 (T)



Post-SFTPRO idle — b2 (units)

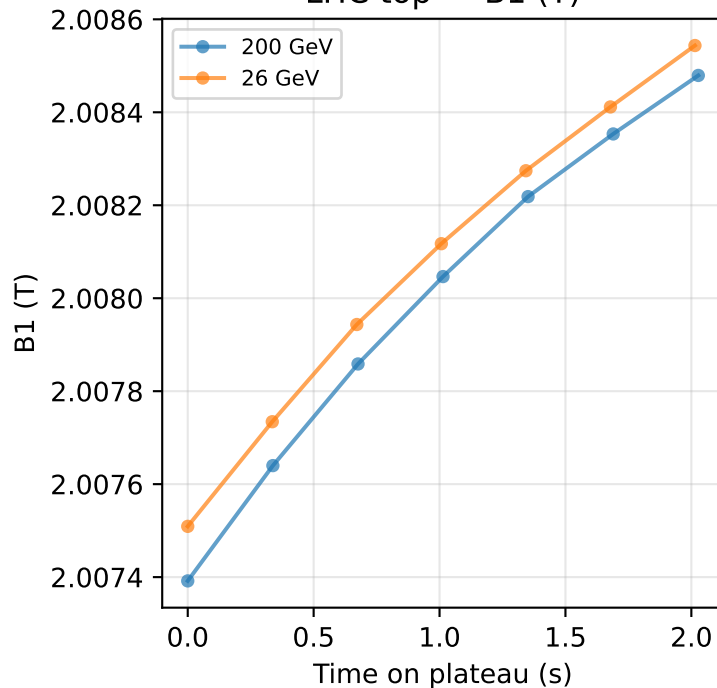


Post-SFTPRO idle — b3 (units)

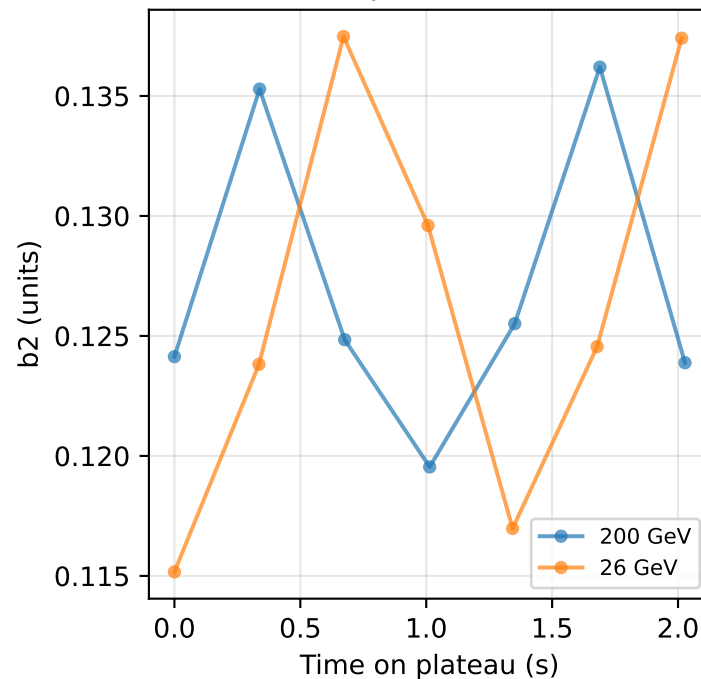


LHC Top Plateau — Turn-by-Turn (body)

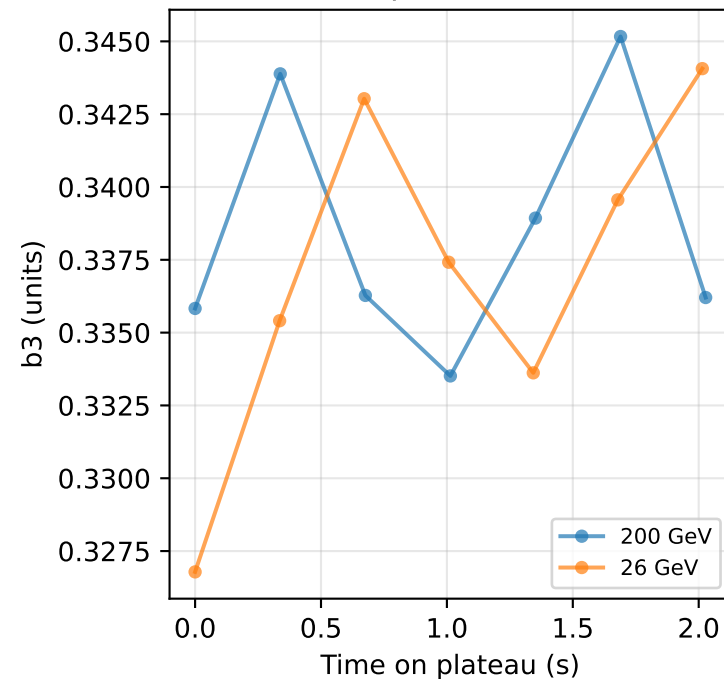
LHC top — B1 (T)



LHC top — b2 (units)

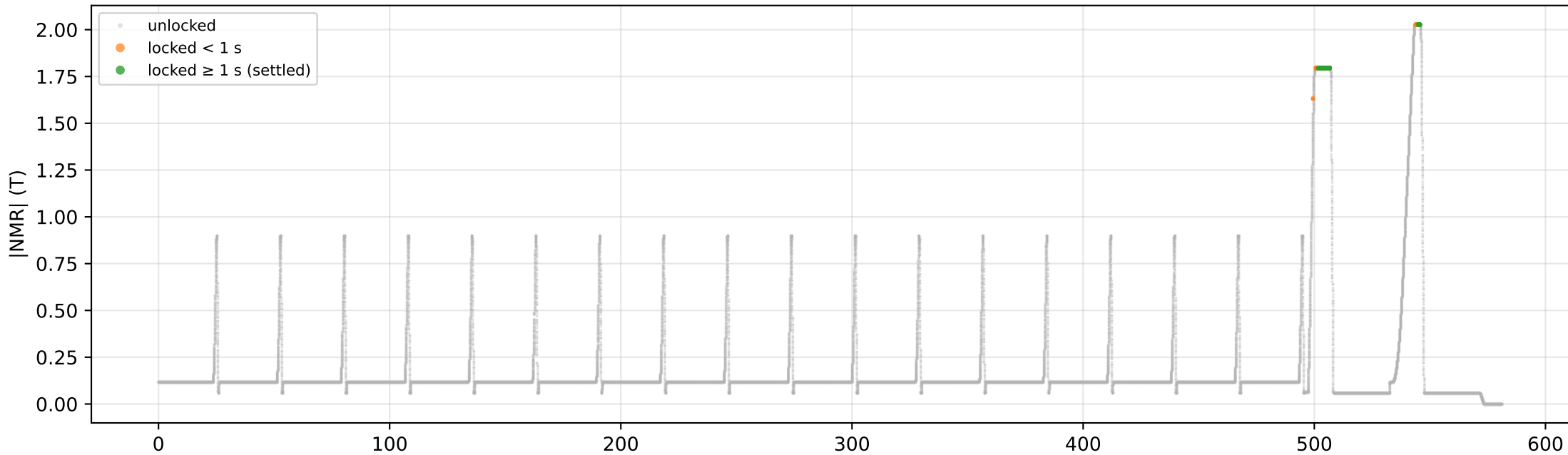


LHC top — b3 (units)

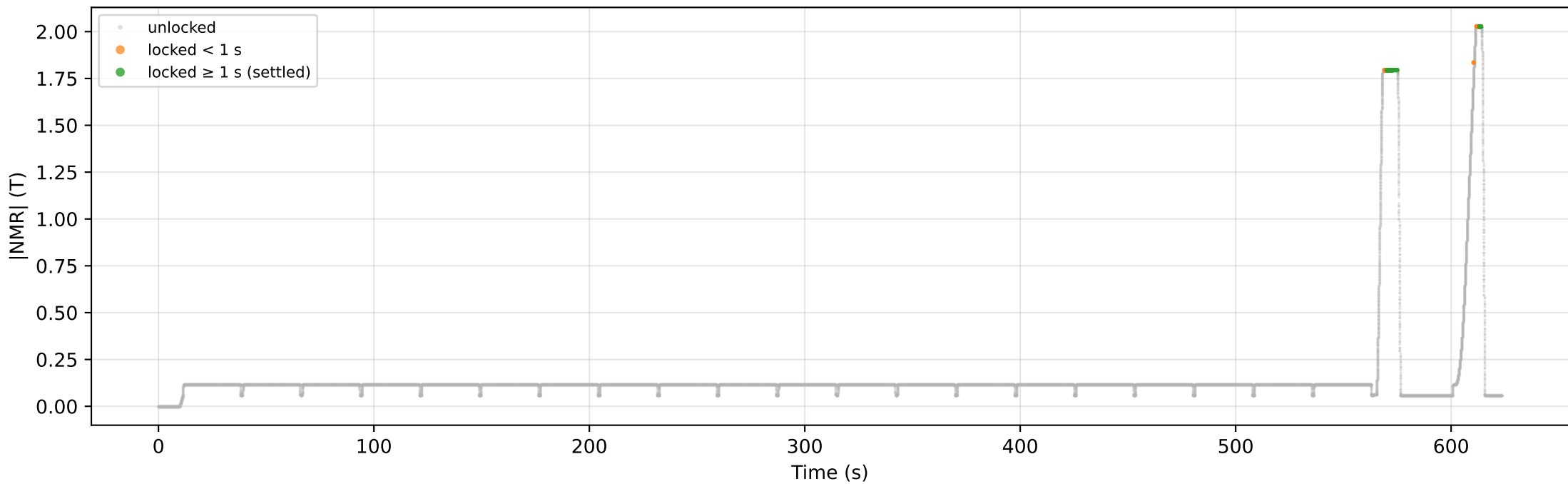


NMR Lock Status — Grey=unlocked, Orange=early, Green=settled

200 GeV — NMR

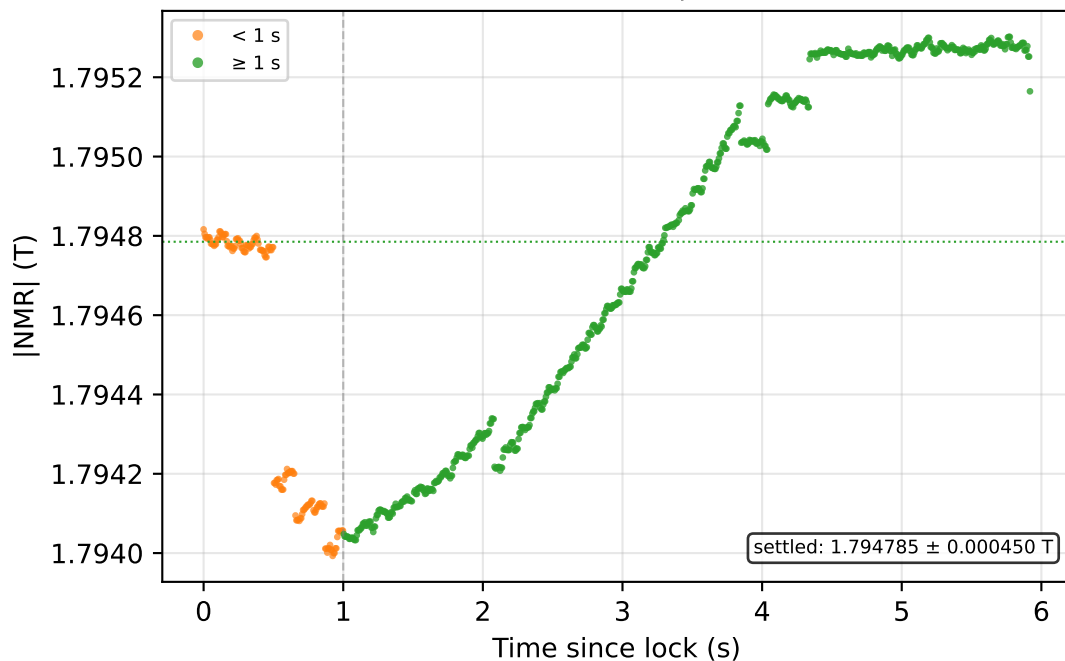


26 GeV — NMR

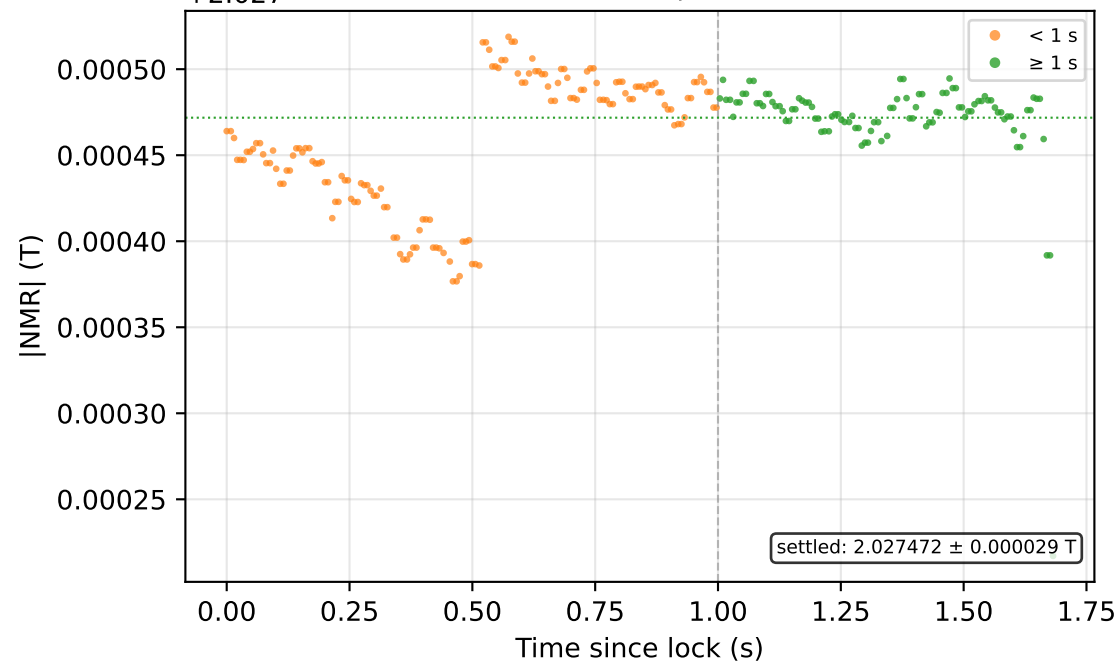


NMR Lock Periods — Zoom

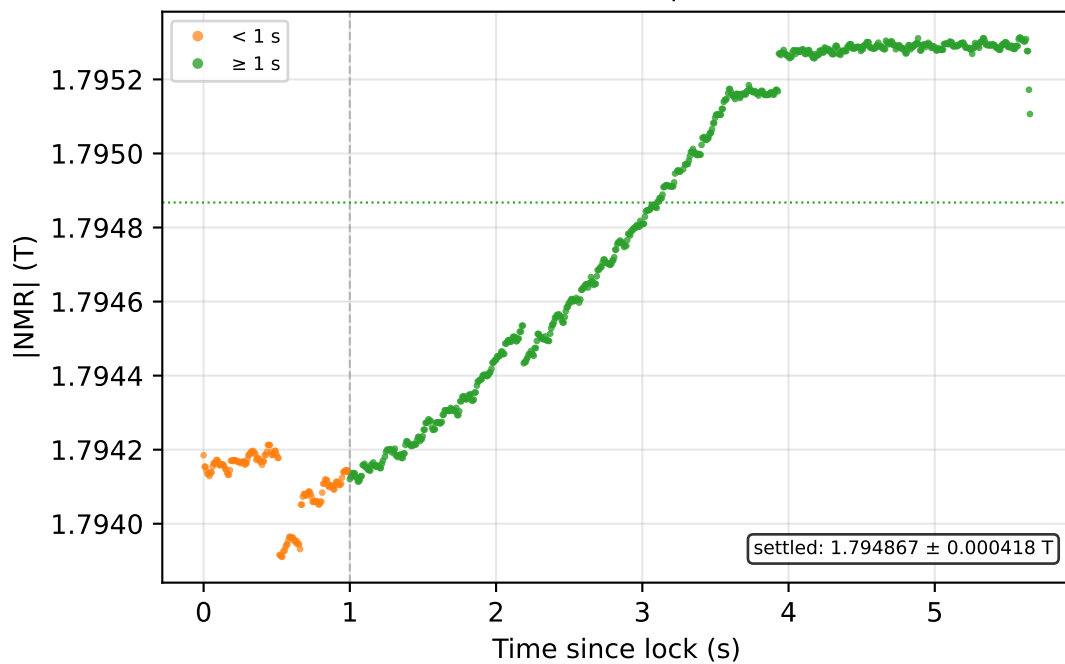
200 GeV — SFTPRO top (5.9 s lock)



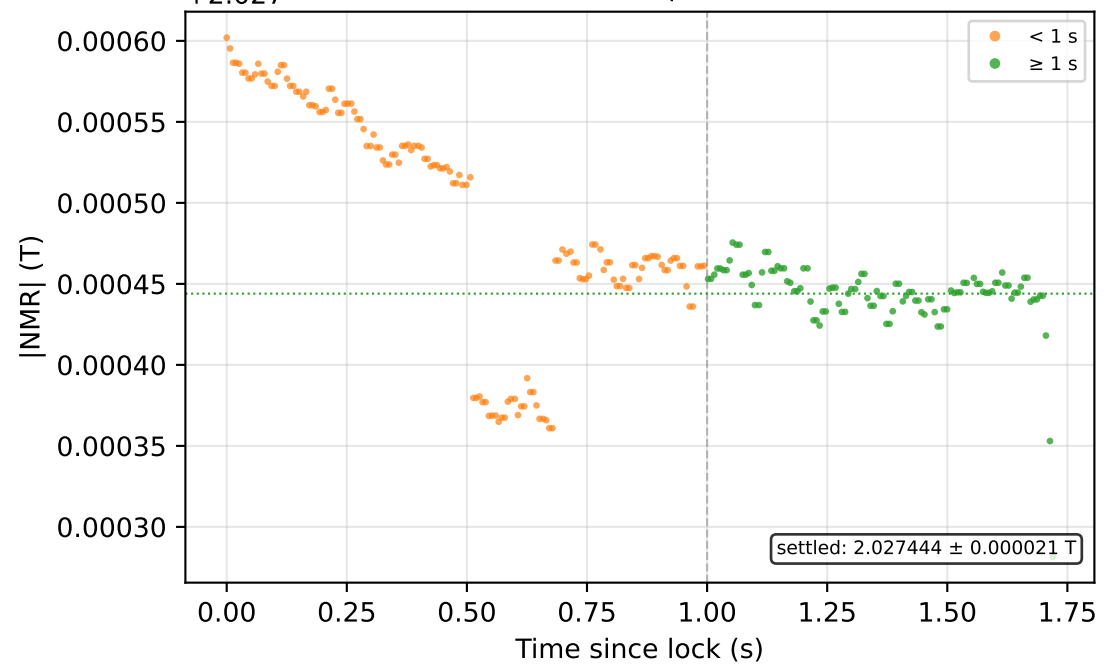
200 GeV — LHC top (1.7 s lock)



26 GeV — SFTPRO top (5.7 s lock)



26 GeV — LHC top (1.7 s lock)



Hysteresis Summary: $\Delta = 26 \text{ GeV}$ (flat MD1) – 200 GeV (full MD1)

Plateau	200 GeV B1 (T)	26 GeV B1 (T)	Δ B1 (μT)	200 GeV b2 (units)	26 GeV b2 (units)	Δ b2 (units)	200 GeV b3 (units)	26 GeV b3 (units)	Δ b3 (units)
SFTPRO top (4816 A)	1.781578	1.781703	125.5	-0.087	-0.086	0.001	0.021	0.023	0.002
SFTPRO→LHC idl (155 A)	0.060503	0.060500	-3.7	0.296	0.278	-0.019	-0.465	-0.457	0.007
LHC injection (302 A)	nan	0.116350	nan	nan	-0.094	nan	nan	-0.084	nan
LHC top (5781 A)	2.008191	2.008258	66.8	0.126	0.129	0.003	0.338	0.340	0.002
Post-LHC idle (155 A)	0.060499	0.060495	-4.3	0.233	0.276	0.043	-0.472	-0.459	0.013
SFTPRO top (NMR)	1.794785	1.794867	82.3	—	—	—	—	—	—
LHC top (NMR)	2.027472	2.027444	-27.8	—	—	—	—	—	—

Body segment only • Encoder offset corrected (equiv. to $-\pi/2$) • Plateau: $|\text{ramp rate}| < 5 \text{ A/s}$
NMR: settled values $\geq 1 \text{ s}$ after lock acquisition

Static Hysteresis Impact — Bar Chart

